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BIOLOGY VALUES CHART

VALUES CHART

UNIT 1

CHP. 1. LIVING WORLD	
Known Species	1.7 – 1.8 Million
Central National Herbarium - Specimens	20,00,000
CHP. 2. BIOLOGICAL CLASSIFICATION	
R.H. Whittaker – Five Kingdom Classification in the year	1969
Size of virus	10 nm to 2000 nm
Adolf Mayer – demonstrated TMV disease is infectious	1886
D.J. Ivanowski – discovered TMV	1982
M.W. Beijerinck – extract could infect healthy plant, called virus venom	1898
W.M. Stanley – Virus could be crystallised	1935
Viroids – discovered by Theodor Diener in the year	1971
No. of nucleotides in PSTV	359 (in free ssRNA)
CHP. 3. PLANT KINGDOM	
No. of flagella in Chlorophyceae	2-8 equal apical
No. of flagella in Rhodophyceae	2, unequal, lateral
Height of kelps	100 m
No. of marine algae spp. used as food	70
Division Bryophyta	960 genera 24,000 spp.
Liverworts – Leafy members – tiny leaf-like appendages	2 rows
Division Pteridophyta	400 genera 10,500 spp.
Term gymnosperm used by Theophrastus	300 BC
Division Gymnospermae	70 genera 725 spp. India - 16 genera 53 spp.
Tallest tree/ Gymnosperm – Sequoia sempervens	366.2 feet/ 111.6 m
Taxodium mucronatum – girth	125 feet
Smallest gymnosperm – Zamia pygma	25 cm
Smallest angiosperm – Wolffia	1 mm
Tallest Angiosperm – Australian Eucalyptus	374 feet/ 114.4 m
CHP. 4. ANIMAL KINGDOM	
No. of known Animal spp.	1.2 Million
No. of Animal Phyla	30 (10 major 20 minor)
No. of arthropod spp.	2/3 rd of all spp. = 11 lakh spp.
No. of mollusc spp.	60,000
Cyclostomata – gill slits	5 to 16 pairs
Chondrichthyes – gill slits	5 to 7 pairs
Osteichthyes – gill slits	4 pairs

UNIT 2

CHP. 5. MORPHOLOGY OF FLOWERING PLANTS	
Root hair	1 -10 mm
Region of cell division	1 mm
Oldest Banyan tree	200 years old 1700 prop roots Crown circumference – 404 m
CHP. 6. ANATOMY OF FLOWERING PLANTS	
No. of vascular bundles in dicot root	2 to 4
No. of vascular bundles in monocot root	6 to many
CHP. 7. (A) ANIMAL TISSUES	
No. of cells in Human body	100 trillion
Types of cells in Human body	200 types
No. of chondrocytes in each lacuna	2-8
CHP. 7. (B) EARTHWORM	
Earthworm segments	(Separate list)
No. of testes	2 PAIRS
Release of worms from cocoon	After 2 weeks
Each cocoon of earthworm	2 – 20 baby worms Avg. of 4
CHP. 7. (C) COCKROACH	
Size of cockroach	¼ to 3 inches (0.6 – 7.6 cm)
Size of <i>Periplaneta americana</i>	34 to 53 mm (L) x 1.5 to 2 cm (B)
No. of sclerites in head	Six
No. of ommatidia in each eye	2000
No. of thoracic segments	3
No. of abdominal segments	11 (11 th reduced and fused with 10 th)
No. of anal cerci (10 th segment)	Pair of 15 jointed
No. of spiracles PAIRS	2 Tho. + 8 Abd.
Alimentary canal length	6-7 cm
Crop	4 th -5 th Abd. Seg.
Chitinous teeth	Six
Hepatic Caeca	6 to 8
Malphigian tubules	100-150
No. of alary muscle PAIRS	12 = 2 Tho. , 10 Abd.
No. of segmental ganglia in nerve cord	9 = 3 Tho., 6 Abd.
Largest Abdominal Ganglion	6 th
Location of 6 th abdominal Ganglion	7 th Abdominal Segment
Location of testes pair	4 th – 6 th Abd. Seg.
Location of Utricular Gland	6 th – 7 th Abd. Seg.
Genital Pouch – Males – Bounded by	Dorsally – 9 th & 10 th Terga Ventrally – 9 th sternum
No. of male gonapophyses	3 PAIRS
Male Gonapophyses arise from	9 th sternum

Location of ovaries PAIR	2 nd – 6 th Abd. Seg.
No. of Ovarian tubules/ ovarioles in EACH ovary	8
Females – boat shaped	7 th Sternum
Genital Pouch – Males – Bounded by	Ventrally – 7 th - 9 th sternum
No. of female gonapophyses	3 PAIRS
Female Gonapophyses arise from	8 th & 9 th sternum
Spermathecae pair	6 th Abd. Seg.
Size of ootheca	3/8 th inch or 8mm
On average, female cockroach produces	9 – 10 oothecae
EACH ootheca contains	14-16 eggs (in 2 rows)
No. of moultings	13
CHP. 7 (D) FROG	
Fore limbs	4 digits
Hind limbs	5 digits
Male frogs – Copulatory pads	1 st digit of forelimb
No. of testes	2 Pairs
No. of Vasa efferentia	10 – 12
No. of ovaries	1 Pair
No. of oviducts	1 Pair
No. of eggs laid by 1 mature female at a time	2500 to 3000

UNIT 3

CHP. 8. CELL: THE UNIT OF LIFE (Only NCERT based)	
Robert Hooke – Cell Discovery	1665
Viruses	0.02 – 0.2 μm
PPLO	0.1 μm
Mycoplasma	0.3 μm
Bacteria (Both NCERT values)	1 – 2/ 3 – 5 μm
Eukaryotic cell	10 – 20 μm
Human RBCS	7 μm
Ribosomes	15 nm x 20 nm
Discovery of electron Microscope	1950s
Plasma Membrane	70 $\text{\AA}$
Membrane of Human RBCs – Protein	52%
Membrane of Human RBCs – Lipid	40%
Discovery of Golgi apparatus – Camillo Golgi	1898
Golgi Cisternae - Diameter	0.5 to 1 μm
Occupancy of plant cell by vacuoles	Upto 90%
Mitochondria - Diameter	0.2 – 1.0 μm (average 0.5 μm)
Mitochondria - Length	1.0 – 4.1 μm
Chloroplast - Diameter	5 – 10 μm
Chloroplast - Length	2 – 4 μm
Number of chloroplasts per cell in Chlamydomonas	1
Number of chloroplasts per cell in mesophyll	20 – 40

Arrangement in Cilia and Flagella	9 (x2) + 2
Arrangement in Centrioles	9 (x3) + 0
Perinuclear space	10 to 50 nm (usually 20 nm)
CHP. 9. BIOMOLECULES	
Element	% Weight
	Earth Crust Human Body
Hydrogen	0.14 0.5
Carbon	0.03 18.5 (2 nd highest)
Oxygen	46.6 (Highest) 65.0 (Highest)
Nitrogen	Very little 3.3
Silicon	27.7 (2 nd Highest) negligible
Component	% of total cellular mass
Water	70-90
Proteins	10-15
Nucleic Acid	5-7
Carbohydrates	3
Lipids	2
Ions	1
Molecular mass of biomicromolecules	18 to 800 Da
Molecular mass of biomacromolecules	Above 1000 Da
Molecular weight of Haemoglobin	68000 Da
Normal blood glucose concentration	4.5 – 5.0 mM
Normal blood hormone concentration	Nanograms/mL
Rate of production of H ₂ CO ₃ in absence of enzymes	200 molecules/hr
Rate of production of H ₂ CO ₃ in presence of carbonic anhydrase	6,00,000 molecules/sec
K _m (Michelis-Menten constant)	10 ⁻¹ to 10 ⁻⁶ M
CHP. 10. CELL CYCLE AND CELL DIVISION	
Yeast – Cell cycle duration	90 minutes
Human cell – Cell cycle duration	24 hours
Human cell – Cell division duration	1 hour
Human cell – Interphase duration	More than 95%

UNIT 4

CHP. 11. TRANSPORT IN PLANTS	
DP of pure water	1236 Atm
Types of aquaporins	8
Percentage of water in watermelon	Over 92%
Herbaceous plants – Dry Weight	10-15%
Transpiration loss in mature corn	3 L/day
Transpiration loss in mustard	Equal to body weight in 5 hrs
OP of pure water	0
Ψ _P of pure water at normal pressure	0
Typical plant cell size	50 μm
Time req. for water to diffuse through a typical plant cell	2.5 sec

Optimum soil temperature for maximum water absorption	20°C to 30°C
1 atm pressure raises water column upto	10.3 m/ 34 feet
Cohesion-Tension-Transpiration Pull theory	1894
Cohesive force of water	350 atm
Amount of absorbed water used	2%
Amount of absorbed water lost in transpiration	98%
Percentage of Cuticular Transpiration	8-10%
Percentage of Stomatal Transpiration	80-90%
Percentage of Lenticular Transpiration	0.1%
pH of guard cells at night (Acc. To proton transport theory)	5.0
Amount of cooling by transpiration	Upto 10 – 15 °C
CHP. 12. MINERAL NUTRITION	
Demonstrations of Hydroponics by Sachs	1860
No. of elements (out of discovered 105) found in different plants	60
Techniques are able to detect minerals even at low concentrations of	10^{-8} g/mL
Amount of macronutrients	In excess of 10 mmole Kg^{-1} of dry matter
Amount of micronutrients	Less than 10 mmole Kg^{-1} of dry matter
Reduction in dry weight of a tissue by ___ % by any mineral is called Mineral toxicity	10%
Amount of energy required for each NH_3 produced	8
CHP. 13. PHOTOSYNTHESIS IN HIGHER PLANTS (Cycles values not added)	
Julius Von Sachs – Evidence of production of glucose	1854
Van Neil experiment	1930
Robert Hill Experiment	1937
Ruben and Kamen Experiment	1941
Dr. Arnon found that H-acceptor is NADP in the year	1954
Length of Porphyrin head in Chl	15 Å
Length of Phytol tail in Chl	20 Å
Phytol tail is attached to ___ th pyrole ring	4 th
Chlorophyll molecule cannot remain in excited state for more than	10^{-9} seconds
Wavelength absorbed by PS – I	700 nm
Wavelength absorbed by PS – II	680 nm
Number of chl-a molecules in LHC	200
Number of chl-b molecules in LHC	Few
Number of carotenoid molecules in LHC	50
Amount of ATP molecules released per electron/ cycle in Cyclic P.P.	Two
Amt. of ATP molecules released/ turn in Non-Cyclic P.P.	One
Quantum requirement	8
Quantum Yield	1/8 i.e. 12.5%

1 NADP uses ____ times more energy to convert to NADPH ₂ than that for ADP to ATP	7.5
Discovery of C ₄ cycle	1954
Nobel Prize to Calvin-Benson	1961
Time required for synthesis of one molecule of Glucose (C ₄ cycle)	90 seconds
Discovery of C ₃ cycle	1970
CO ₂ Loss in Photorespiration	25%
Light saturation	10% of full sunlight
Optimum temperature for C ₃ plants	20-25°C
Optimum temperature for C ₄ plants	30-40°C
CO ₂ compensation point for C ₃ plants	0-10 ppm
CO ₂ compensation point for C ₄ plants	25-100 ppm
CO ₂ conc. upto which photosynthesis rate is increased	0.05%
Optimum CO ₂ concentration	0.03 – 0.04%
CO ₂ saturation for C ₃ plants	360 µL ⁻¹ (ppm)
CO ₂ saturation for C ₄ plants	450 µL ⁻¹ (ppm)
CHP. 14. RESPIRATION IN PLANTS (Cycles values not added)	
Amount of energy released during oxidation of one molecule of glucose	686 Kcal
Amount of above energy stored as chemical energy	40%
Amount of above energy released as heat energy	60%
Energy stored in ATP in 1 st phosphate bond	3.4 Kcal
Energy stored in ATP in 2 nd phosphate bond	7.3 Kcal
Energy stored in ATP in 3 rd phosphate bond	7.3 Kcal
Energy released during ATP to ADP + iP	7.3 Kcal
RQ (CAM)	0
RQ (Fat Tripalmitin C ₅₁ H ₉₈ O ₆)	0.7
RQ (Protein)	0.9
RQ (Carbohydrate)	1
RQ (Citric Acid)	1.3
RQ (Malic Acid)	1.33
RQ (Oxalic Acid)	4
RQ (Anaerobic respiration)	Infinity
RQ (Germinating fatty seeds E.g. Castor)	<1
RQ [Formation of fatty seeds (Maturing fatty seeds)]	>1
CHP. 15 PLANT GROWTH AND DEVELOPMENT	
Stratification	Warm – 15 to 20°C Cold – 0 to 10°C
Growth rate in single maize root apical meristem	17500 new cells/hour
Increases in size of watermelon cells	3,50,000 times
Experimental proof for auxins by F.W.Went	1928
Sugarcane yield after using Gibberellins	20 tonnes/acre
Gane reported about ethylene	1934
Garner Allard Photoperiod Observation	1920
Cajlachjan named florigen	1935

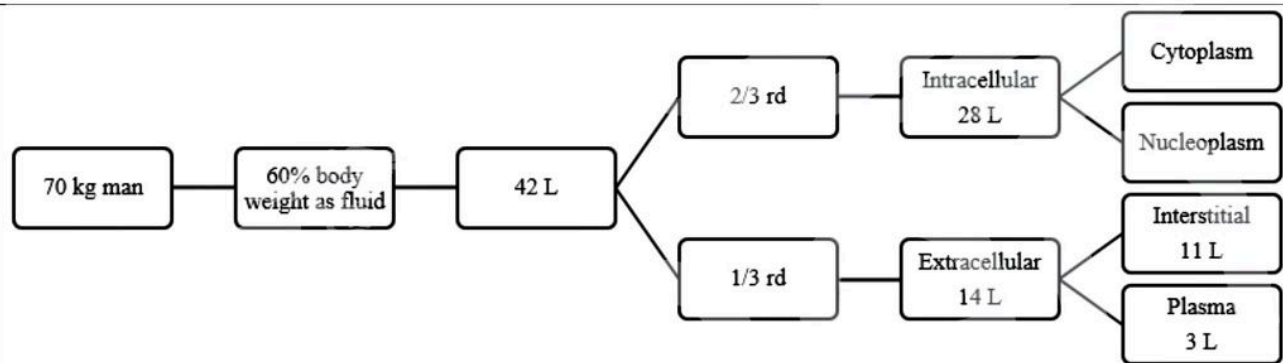
Klippiart reported about Vernalisation	1857
Lyсенко gave the term Vernalisation	1928

UNIT 5

CHP. 16. DIGESTION AND ABSORPTION						
Human body – Percentage of water			65% water (60-80 % body weight)			
Percentage of Carbohydrates in balance diet			60-70%			
Percentage of Fats in balance diet			20-30%			
Percentage of Proteins in balance diet			10-20%			
Energy requirement - Older people			2500 cal/day			
Energy requirement - Young active/ Growing Children			3700 cal/day			
Length of Alimentary canal			8 m			
Length of oesophagus			25 cm			
Length of stomach			25 cm			
Dimension of small intestine			6 m long, 2.5 cm broad			
Length of Duodenum			25 cm			
Length of Jejunum			2.5 m			
Length of Ileum			3.5 m			
Dimensions of Large Intestine			1.5 m long, lumen – 6.5 cm broad			
Mouth pH			6.8			
Stomach pH			1.8			
Intestine pH –			7.6			
Liver Weight			1.5 kg			
Liver lobes fractions			5/6 Right, 1/6 Left			
	(kcal/g)		Gross Cal. Val.		Physiological Val.	
	Fats		9.45		9.0	
	Proteins		5.65		4.0	
	Carbs		4.1		4.0	
Adults – Protein Requirement			1 g/ kg body weight/ day			
Children – Protein Requirement			2 g/ kg body weight/ day			
CHP. 17. BREATHING AND EXCHANGE OF GASES						
Length of Pharynx			12 cm			
Dimensions of trachea			12 cm (L) x 2.5 cm (w)			
Number of cartilaginous rings on trachea			16-20			
Trachea divides at the level of			5 th thoracic vertebra			
Weight and number of lobes of right lung			625 gms – 3 lobes			
Weight and number of lobes of left lung			570 gms – 2 lobes			
	Gas	Atmo. Air	Alveoli	Oxy Blood	Deoxy Blood	Tissues
	O ₂	159	104	95	40	40
	CO ₂	0.3	40	40	45	45
Normal breathing rate			16-18 / min			
Breathing rate during exercise			Upto 80 / min			
Infants breathing rate			44 /min			
TV			500 mL/ breath			

	6000-8000 ml/ min
IRV	2500-3000 mL/ min
ERV	1000-1100 mL/min
RV	1100-1200 mL/min
Solubility of CO ₂ = Solubility of O ₂ x ____ times	20-25
O ₂ Transport	97% RBC, 3% Plasma (dissolved)
CO ₂ Transport	70% Bicarb, 23% RBC, 7% Plasma (dissolved)
Chloride shift maintains equilibrium pH	7.4
Amount of O ₂ transported	5 ml/100 ml blood
Amount of CO ₂ transported	4 ml/100 ml blood
Amount of CO ₂ transported by bicarbonate	2.5 ml/100 ml blood
Inspiration time	2 sec
Expiration time	3 sec

CHP. 18. BODY FLUIDS AND CIRCULATION



Transcellular fluid	1 to 2 L
Discovery of closed circulatory system	1628
pH of blood	7.4
Blood volume	Male – 5 to 6 L Female – 4 to 5 L
Percentage of plasma in blood	55%
Percentage of blood corpuscles in blood	45%
Percentage of water in plasma	90-92%
Percentage of solutes in plasma	3%
Percentage of proteins in solutes	7%
Size of RBCs	7-8 µm diameter 2 µm thickness
Total RBC count	5 – 5.5 million/ cu.mm. of blood
Life span of RBC	100 -120 days
Normal haemoglobin count	Male – 13-18 mg/dL Female – 11.5-16.5 mg/dL
Size of WBCs	8 to 15 µm
Total WBC count	6000-8000
Life span of WBCs	10 to 13 days
Percentage of total WBCs as neutrophils	60-65%
Percentage of total WBCs as eosinophils	2-3%

Percentage of total WBCs as basophils		0.5-1%	
Percentage of total WBCs as lymphocytes		20-25%	
Percentage of total WBCs as monocytes		6-8%	
Size of platelets		2.5-5 μ m	
Total platelet count		1.5-3.5 million/ cu.mm. of blood	
Life span of platelets		5 to 10 days	
ABO system of blood group discovered in		1900	
AB blood groups discovered in		1902	
Bleeding time		3-6 min	
Clotting time		3-8 min	
No. of blood clotting factors		13	
Dimensions of heart		12 cm x 9 cm	
Weight of heart		250 – 300 grams	
Life of artificial pacemaker		3 -7 years	
Duration of Cardiac Cycle		0.8 Second	
	Atrial Systole	0.1 Second	
	Ventricular Systole	0.3 Second	
	Joint Diastole	0.4 Second	
Stroke Volume		70 mL	
Heart rate, Pulse rate		72/ min	
Cardiac output		5040 L/ min	
Tachycardia heart rate		>100 /min	
Bradycardia heart rate		< 60 /min	
Normal systolic blood pressure		120 mm Hg	
Normal diastolic blood pressure		80 mm Hg	
Hypertension systolic blood pressure		>140 mm Hg	
Hypertension diastolic blood pressure		>90 mm Hg	
CHP. 19. EXCRETORY PRODUCTS AND THEIR ELIMINATIONS			
Water required for excretion as ammonia		300-500 mL/gm	
Water required for excretion as urea		50 mL/gm	
Water required for excretion as uric acid		10 mL/gm	
Location of kidney in terms of vertebrae		12 th thoracic to 3 rd lumbar	
Dimensions of kidney		10-12 cm (L) x 5-7 cm (W) x 2-3 cm (T)	
Weight of kidney		120-170 gms	
Number of renal pyramids		6 – 20	
Length of each ureter		40 cm	
Storage capacity of urinary bladder		500 mL to 1 L	
Length of Urethra (males)		20 cm	

Length of Urethra (females)	4 cm
No. of nephron in EACH kidney	1 to 1.2 million
Dimensions of each nephron (TOTAL length x diameter)	3 cm x 20-60 μ m
Percentage of cortical nephrons	85% (75-85%)
Percentage of juxtamedullary nephrons	15% (15-25%)
<pre> graph LR A[Cardiac Output 5 L] --> B[20 - 25% to kidney Renal Blood Flow 1100-1200 mL/min] B --> C[55% Plasma Renal Plasma Flow 650 mL] C --> D[20% Filtration GFR 125 mL/ min 180 L/ day] </pre>	
Filtration fraction = GFR/RPF	$1/5 = 20\%$
Glomerular hydrostatic pressure (GHP)	(+) 55 mm Hg
Blood colloid osmotic pressure (BCOP)	(-) 30 mm Hg
Capsular Hydrostatic Pressure (CHP)	(-) 15 mm Hg
Urine released per day	1 – 1.5 L
Percentage reabsorption	99%
Percentage of electrolytes selectively reabsorbed in PCT	70-80%
% in urine: Water	95%
% in urine: Urea	2.6%
% in urine: Electrolytes	2%
% in urine: Uric acid, Creatinine, Ammonia, Hippuric Acid	0.3% EACH
Amount of urea excreted per day	25-30 gm
pH of urine	6.0
Normal percentage of urea in blood	0.01 to 0.03%
Uremic percentage of urea in blood	>0.05%
Streptococcal glomerulonephritis commonly seen in children of age	6 to 16
Oliguria measure in adults	<400 mL/day
Oliguria measure in children	<0.5 mL/kg/hr
Oliguria measure in infants	<1.0 mL/kg/hr
CHP. 20. LOCOMOTION AND MOVEMENT	
No. of muscles in humans	640
No. of bones → separate chart	
No. of sacral vertebrae	5 (fused)
No. of coccygeal vertebrae	4 (fused)
Length of sternum	15-17 cm
No. of true ribs	7 pairs (1 st to 7 th)
No. of false ribs	3 pairs (8 th to 10 th)
No. of floating ribs	2 pairs (11 th to 12 th)
CHP. 21. NEURAL CONTROL AND CO-ORDINATION	
Na-K pump: No. of Na ⁺ efflux	3
Na-K pump: No. of K ⁺ influx	2
Speed of nerve impulse in Man	130 m/s
Speed of nerve impulse in frog	30 m/s

Development time of brain	70-80% - 2 years Complete – 6 years
Development time of spinal cord	4 to 5 years
Weight of adult brain (Male/ NCERT)	1400 gm
Weight of adult brain (Female)	1250 gm
pH of CSF	7.28-7.33
Rate of CSF production	0.5 mL/ min = 720 mL/ day
Constant amount of CSF	120 mL
Reduction of weight of brain due to buoyancy provided by CSF	1.4 kg to 0.18 kg
Percentage of cerebrum in brain	80-85%
Percentage of thalamus in diencephalon	80%
Which and how many cranial nerves arise from medulla oblongata?	7 th to 12 th (6)
Percentage of cerebellum in brain	11%
Length of spinal cord	45 cm
Length of filum terminale	20 cm
Weight of spinal cord	35 gm
Which cranial nerves are optic sensory	III, IV, VI
Number of spinal nerves	31 pairs
Number of cervical spinal nerves	8 pairs
Number of thoracic spinal nerves	12 pairs
Number of lumbar spinal nerves	5 pairs
Number of sacral spinal nerves	5 pairs
Number of coccygeal nerves	1 pair
No. of ganglia in Sympathetic ANS	22
Number of muscles responsible for eye rotation	6
Number of lacrimal ducts	6-12
Number of layers in eye wall	3
CHP. 22. CHEMICAL CONTROL AND CO-ORDINATION	
First hormone secretin discovered in year	1902
Weight of pituitary gland	0.5 g
Percentage of anterior pituitary in pituitary	75%
Percentage of chromophils in ant. pituitary	50%
Percentage of chromophobes in ant. pituitary	50%
Percentage of acidophils in ant. pituitary	43%
Percentage of basophils in ant. pituitary	7%
Percentage of posterior pituitary in pituitary	25%
Weight of thyroid gland	25-30 grams
Dimensions of thyroid gland	5 cm x 3 cm
No. of lobes of thyroid gland	2
Location of isthmus (acc. To trachea)	2 nd -4 th tracheal cartilage
TOTAL number of thyroid follicles	3 million

Parathyroid glands are made from epithelium of _____ branchial pouches	3 rd & 4 th
Normal level of blood Ca ⁺⁺	12 mg/dL
Percentage of cortical portion in adrenal gland	80-90%
Percentage of medullary portion in adrenal gland	10-20%
Number of hormones synthesized in adrenal cortex	40-50
Number of active hormones out of above	7-8
Percentage of acini in pancreas	99%
No. of islets of Langerhans	1-2 million
Percentage of endocrine part in pancreas	1-2%
Percentage of α cells in pancreas	25%
Percentage of β cells in pancreas	60-65%
Percentage of γ cells in pancreas	10%
Total no. of amino acids in insulin	21 (A) + 30 (B) = 51
No. of chains in insulin	2
No. of amino acids in C chain in insulin	31
No. of disulphide bridges in insulin	2 between the chains + 1 along A chain = 3
World diabetes day	14 November
Insulin Shock – blood glucose level	<40 mg/dL

UNIT 6

CHP. 1. REPRODUCTION IN ORGANISMS

Butterfly = 1-2 weeks	Elephant = 60-90 years
Fruit fly- = 20-30 days	Parrot = 140 years
Crow = 15 years	Tortoise = 100-150 years
Cow = 15-25 years	Rose = 5-7 years
Dog = 20-30 years	Rice plant = 3-4months
Crocodile = 60 years	Banana tree = 20-25 years
Horse = 62 years	Banyan tree = 200-300 years

Organism	Chromosome no. in meiocyte (2n)	Organism	Chromosome no. in meiocyte (2n)
Human beings	46	Fern ophioglossum	1260
House fly	12	Apple	34
Rat	42	Rice	24
Dog	78	Maize	20
Cat	38	Potato	48
Fruit fly	8	Butterfly	380
Yeast	16	Onion	32

Length of scion	6-12 inches
Time taken by Marigold, rice, wheat to flower	3 months
Time taken by coconut to flower	7 years

Time taken by Mango to flower	3-6 years
Time taken by Bamboo to flower	50-100 years
Neelakuranji flowers once in	12 years
Last flowering of Neelakuranji	September-October 2006 (NCERT)

CHP. 2. SEXUAL REPRODUCTION IN FLOWERING PLANTS

Number of middle layers	1 to 3
Size of pollen grain	25-50 μm
Number of germs pores in monocots pollen	1
Number of germs pores in dicots pollen	3
Percentage of angiosperms – pollen released at 2 celled stage	60%
Percentage of angiosperms – pollen released at 3 celled stage	40%
Temperature of liquid nitrogen	– 196°C
Pollen grain observation – concentration of sugar solution	10%
Pollen grain observation – time	15-30 minutes
Viability of pollen grains of rice and wheat	30 minutes
Viability of pollen grains of Rosaceae, Solanaceae, Leguminosae	Few months
Polygonum type embryo sac found in ____ % Angiosperms	70%
Pollination discovered in the year	1694
% of insect pollination by bees	70-80%
Discovery of Double fertilization	1898
Viability/ dormancy of seed of Lupinus arcticus	10,000 years
Viability of seed of Phoenix dactylifera	2000 years

CHP. 3. HUMAN REPRODUCTION

Difference in temperature of scrotum (lower) and abdomen	2-2.5°C
Dimensions of testes	4-5 cm (L) x 2-3 cm (W) x 3 cm (T)
EACH testis divided into ____ lobules	250
No. of seminiferous tubules in a lobule	1-3
Length of epididymis	6 m. (20 feet)
Length of Urethra (males)	20 cm
Length of Urethra (females)	4 cm
Percentage accountancy of semen by seminal fluid	60%
Percentage accountancy of semen by prostatic fluid	30%
Dimension of ovaries	2-4 cm (L) x 2 cm (W)
Length of fallopian tube	10-12 cm
Dimensions of uterus	3 x 2 x 1 inches
Length of vagina	3½ inches
Number of mammary lobes in EACH breast	15-20
Length of sperm	0.055 mm (60 μm)
Viability of sperm (survives for)	72 hours (3 days)
Sperm can fertilize ovum in first	12 to 14 hours
Normal semen sperm count during a coitus (3-5 mL semen)	200 to 300 million
Percentage of sperms having normal shape for fertility	60%
Percentage of sperms having vigorous motility for fertility	40%

No. of oogonia in EACH foetal ovary (by 25 weeks) prior to birth	2 million
Number of primary follicles left in EACH ovary at puberty	60,000 to 80,000
Approximate reproductive age for female	15-45 years old
No. of layers of granulosa cells in Theca interna upper $\frac{3}{4}$ th area	2
No. of layers of granulosa cells in Theca interna upper $\frac{1}{4}$ th area	6-7
Size of ovum	0.1 mm or 100 μ m
Ovum survives for	2 days
Age of Menarch	12-14 years old
Age of menopause	45-50 years old
Usual menstrual cycle time period	28 days
Thickness of endometrium at end of menstrual phase	0.5 to 1 mm
Thickness of endometrium at end of proliferative phase	2 – 3 mm
Thickness of endometrium at end of luteal phase	4 – 5 mm
Time of completion of cleavage after fertilization	3 days
Time of completion of blastulation after fertilization	5 days
Time of implantation after fertilization	7-8 days
Time of completion of gastrulation after fertilization	15 days
Time of fertilization after ovulation (within)	24 hours
Time of 1 st cleavage after fertilization	30 hours
Time of 2 nd cleavage after fertilization	60 hours
Time of 3 rd cleavage after fertilization	72 hours
Size of blastula	0.15 to 0.30 mm
Period of human pregnancy (from fertilization)	266 days
Period of human pregnancy (from start of last menstrual cycle)	280 days
Period of dogs pregnancy	58-68 days
Period of elephants pregnancy (longest)	18-22 months
Period of cats pregnancy	64-67 days
Length of foetus during first trimester	5 cm
All major structures of adult are present in rudimentary form in foetus by end of	8 weeks
Length of foetus during second trimester	30 cm
Length of foetus during third trimester	50 cm
Weight of foetus during third trimester	3-4 kg
First foetal movements are seen during	5 th month
Human placenta dimensions	4 cm thick x 18 cm diameter
Average weight of placenta	1/6 to 1/7 that of foetus
Length of foetus during parturition	53 cm
Weight of foetus during parturition	2.7-4.5 kg
Time of first stage (dilation) of parturition	12 hours
Time of second stage (expulsion) of parturition	20 minutes to 1 hour
Time of third stage (placental) of parturition	10 – 45 minutes
CHP. 4. REPRODUCTIVE HEALTH	
Family planning programmes in India	1951
Marriageable age for females	18 years
Marriageable age for males	21 years

World population in 1900	2 billion
World population in 2000	6 billion
Indian population during independence	350 million
Indian population by 2000	1 billion
Indian population reached 1 billion	May 2000
Current population growth rate of India	1.7%
Number of years for Indian population to double by current growth rate	33 years
Periodic abstinence – Avoid coitus during	10 to 17 th day of M.C.
Oral contraceptives are to be taken for a period of	21 days
Break in oral contraceptives	7 days
Implants prevent pregnancy for	3 to 4 years
Emergency contraceptives to be given within	72 hours of coitus
No. of MTPs per year (1/5 th of total conceived pregnancies)	45 to 50 million
GOI legalized MTP in	1971
MTPs are safe upto	12 weeks
STDs are high among age group	15-24 years
Worldwide cases of syphilis	50 million
Worldwide cases of gonorrhea	150 million
Incubation period for syphilis	3-4 weeks
Incubation period for gonorrhea	Males – 2-14 days Females – 7-21 days
1 st test tube baby in the world born on	25 th July 1978

UNIT 7

CHP. 5 PRINCIPLES OF VARIATION AND INHERITANCE	
William Bateson coined the term genetics	1906
Years in which Mendel performed expt.	1856-1863
No. of years for which Mendel performed expt.	7
Mendel born on	22 nd July 1822
Mendel presented his paper in the year	1865
Mendel's papers were published in journal	1886
Mendel died in the year	1884
Rediscovery of Mendel's papers	1900
Mendel's paper were republished in Flora	1901
Area of Mendel's plots	70m x 7m
Phenotypic ratio of F2 gen – Monohybrid	3:1
Genotypic ratio of F2 gen – Monohybrid	1:2:1
Phenotypic ratio of F2 gen – Dihybrid	9:3:3:1
Genotypic ratio of F2 gen – Dihybrid	1:2:2:4:1:2:1:2:1
Phenotypic ratio for Incomplete/ Co-dominance	1:2:1
Pleiotropic gene : Lethal in double recessive	2:1
Polygenic Inheritance: 2 genes	1:4:6:4:1
Polygenic Inheritance: 3 genes	1:6:15:20:15:6:1
Additive genes	9:6:1

Duplicate genes	15:1
Inhibitory gene	13:3
Dominant epistasis / supplementary genes	12:3:1
Recessive epistasis	9:3:4
Complementary genes	9:7
Morgan's expt. – body colour and eye colour – Parental combination	98.7%
Morgan's expt. – body colour and eye colour – Recombinants	1.3%
Morgan's expt. – body colour and wing size – Parental combination	62.8%
Morgan's expt. – body colour and wing size – Recombinant	37.2%
Discovery of X body	1891
Thalassemia: Gene for α chain on _____ chromosome	16
Thalassemia: Gene for β chain on _____ chromosome	11
Discovery of Thalassemia	1925
Sickle cell anemia: substitution at _____ position of β chain	6 th
Down's Syndrome	21 trisomy
Frequency of Down's syndrome	1 in 700
Edward's Syndrome	18 trisomy
Patau's Syndrome	13 trisomy
Cat cry syndrome: Deletion of short arm of chromosome	5
Frequency of Klinefelter's syndrome: _____ male birth	1 in 850
Frequency of Turner's syndrome: _____ female birth	1 in 3000
Frequency of Jacob's syndrome: _____ female birth	1 in 1000
CHP. 6. MOLECULAR BASIS OF INHERITANCE	
No. of bases/ nucleotide: ϕ x 174 bacteriophage	5386
No. of bases/ nucleotide: λ bacteriophage	48502
No. of bases/ nucleotide: Escherichia coli	4.6×10^6 bp
No. of bases/ nucleotide: Human DNA	3.3×10^9 bp
Watson-Crick DNA model - Proposal	1953
Nobel prize to Watson-Crick-Wilkins	1962
Each step (between two consecutive base pairs) in DNA	36° , $3.4 \text{ \AA}$ (0.34 nm)
10 bp (Pitch)	360° , $34 \text{ \AA}$ (3.4 nm)
No. of Hydrogen bond between A and T/U	3
No. of Hydrogen bond between G and C	2
Major grooves length	$22 \text{ \AA}$
Minor grooves length	$12 \text{ \AA}$
Length of DNA in average mammalian cell	2.2 meters
Length of DNA in E. coli	1.36 mm
Length of DNA on histone octamer	146 bp
Length of DNA on H1 histone (linker DNA)	54 bp
Length of DNA per nucleosome	200 bp
Diameter of DNA in nucleosome fiber	10 nm
Diameter of DNA in solenoid fiber	30 nm
Diameter of DNA in chromatin fiber	200 – 300 nm
Diameter of DNA in chromatid	700 nm

Year of Griffith experiment	1928
Year of Avery, Macleod and McCarty experiment	1944
Year of Hershey and Chase experiment	1952
m-RNA: % of total cellular RNA	3-5%
Molecular weight of m-RNA	5,00,000
t-RNA: % of total cellular RNA	10-20%
Molecular weight of t-RNA	23,000 – 30,000
Length of t-RNA	73-93 nucleotides
r-RNA: % of total cellular RNA	80%
Molecular weight of r-RNA	40,000 – 1,00,000
Proposal of DNA replication mechanism by Watson and Crick	1953
Messelson and Stahl proof for semi-conservative DNA replication	1958
Taylor's proof for semi-conservative DNA replication	1958
Rate of DNA replication in E. coli	2000 bp/sec
Year of Beadle and Tatum experiment on Neurospora crassa	1941
Vernon Ingram: One gene one polypeptide hypothesis: Year	1957
Time required for E. coli to divide	18-20 min
No. of adenylate residues at 3' end during tailing	200-300
George Gamow: Codon	1954
Direct evidence for codon by crick	1961
Nobel prize to Nirenberg, Matthaei & Khurana	1968
Number of essential amino acids	21
Number of codons	64
No. of stop codons (considering 20 amino acids)	3
Wobble hypothesis	1966
Percentage of r-RNA in ribosomes	65%
Percentage of proteins in ribosomes	35%
Discovery of operon	1961
No. of genes (grouped in operons) in E. coli	260
No. of operons in E. coli	75
CHP. 7. EVOLUTION (only NCERT)	
Formation of earth	4.5 bya
Life appeared on earth	500 mya
Miller's experiment	1953
Temperature in Miller's experiment	800°C
First cellular form	200 mya
Age of earth according to special creation theory	4000 years
Industrial Melanism: Year in which light moths were more	1850s
Industrial Melanism: Year in which dark moths were more	1920
Invertebrates were active at	500 mya
Jawless fish evolved	350 mya
Sea weeds and few plants existed	320 mya
Fish with stout and strong fins could move on land and go back to water	350 mya
Fish like reptiles evolved	200 mya

Height of Tyrannosaurus rex	20 feet
Dinosaur suddenly disappeared from earth	65 mya
Australopithecus period	3-4 mya
Height of Australopithecus	<4 feet
Brain capacities of Homo habilis	650-800 cc
Homo erectus fossils discovered in Java	1891
Homo erectus period	1.5 mya
Homo erectus brain capacity	900 cc
Neanderthal man brain capacity	1400 cc
Neanderthal man period	1,00,000-40,000 yr. ago
Ice age (Modern humans arose)	7,500 – 10,000 years ago
Pre-historic cave art developed	18,000 years ago
Agriculture came around	10,000 years ago

(other values of human evolution in human evolution chart)

UNIT 8

CHP. 8. HUMAN HEALTH AND DISEASE	
No. of sporozoites released from each oocyst	10,000
Malarial fever and sweating lasts for	Four to six hours
Frequency of fever in malaria by <i>P. falciparum</i>	Every 3 days
Frequency of fever in malaria by <i>P. ovale</i> and <i>P. vivax</i>	Every 2 days
Number of known filarial nematodes	Nine
Number of developmental stages in filarial worm	Five
Round worm adults live upto	1-2 years
First appearance of ascaris eggs in stools	After 60-70 days
Days required to show symptoms after infection	4-16 days
Name typhoid was given by Louis in year	1829
Incubation period for typhoid	10-20 days
Typhoid fever temperature	39-40°C (104°F)
Typhidot has to be performed within how many days?	7-14 days
Incubation period of Pneumonia	1-3 days
Common Cold: Beginning of symptoms after ____ hours of exposure	16
Frequency of common cold in children	12 /year
Frequency of common cold in adults	6 /year
Common cold usually lasts for	7-10 days, even upto 3 weeks
Rate of production of antibodies by B cells	2000 Ab molecules/ sec/ cell
Production of antibodies by each cell occurs for _____ days until plasma cell dies	4 to 5
Percentage of IgG	70-75%
Percentage of IgM	5-10%
Percentage of IgA	10%
Percentage of IgD	<1%

Percentage of IgE	<1%
AIDS was first reported in	1981
No. of deaths due to AIDS	25 Million
Size of HIV	100 nm
Incubation period of HIV	2-10 years
Window period of HIV	First 2-6 months
CD4 T-cell count during AIDS	200 cells/mm ³
Amount of WBCs during leukemia	2,00,000 to 10,00,000 cells/mm ³
Adolescence	13 to 19 years
CHP. 9. STRATEGIES FOR ENHANCEMENT IN FOOD PRODUCTION	
Percentage of rural manpower dependent on agriculture	80%
Percentage of world's livestock in India and China	70%
Percentage of world farm produce by India and China	25%
Inbreeding involves mating of closely related individuals within same breed for	4-6 generations
In each cycle of MOET, _____ eggs mature simultaneously	6 to 8
Fertilized eggs at _____ stage are recovered	8-32 celled
India contributes _____ % of total lac produced in the world	85
Recorded evidence of plant breeding	9000-11000 years ago
Agriculture accounts for _____ % of India's GDP	33
Agriculture employs _____ % of India's population	62
Wheat production increase during 1960 to 2000	11 million tonnes to 75 million tonnes
Rice production increase during 1960 to 2000	35 million tonnes to 89.5 million tonnes
Sonalika and Kalyansona introduced in the year	1963
IR-8 and Taichung Native 1 introduced in the year	1966
Number of people in the world who do not get adequate food to meet daily nutritional requirements	>840 million
Number of people suffering from protein, vitamin, etc. deficiency	3 billion
Amount of grains to produce 1 kg of meat	3-10 kg
250 kg cow produces _____ protein/day	200 g
250 g of Methylophilus methylotrophus produces _____ protein/day	25 tonnes
pH of medium for tissue culture	5 to 5.8
Rate of agitation in suspension culture	100-250 rpm
CHP. 10. MICROBES IN HUMAN WELFARE	
Discovery of Penicillin	1929
Nobel prize for penicillin	1945
Number of types of Gas isolated from G. fujikoroii	15
Height of concrete tank – biogas plant	10-15 feet deep
Percentage of CH ₄ in biogas	50-80%
Percentage of CO ₂ in biogas	15-45%
Number of known types of NPV	125

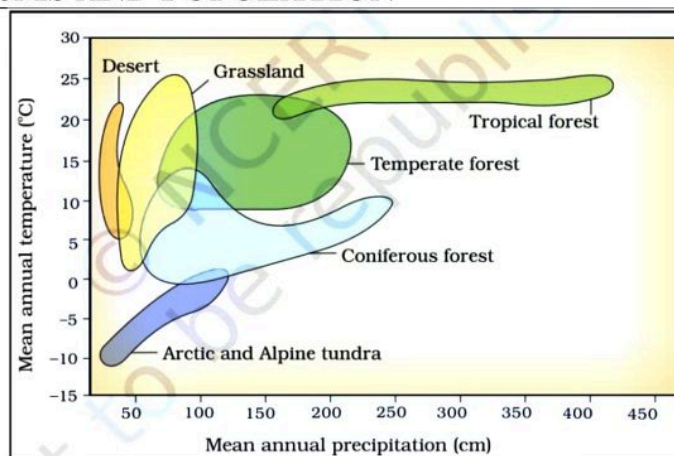
UNIT 9

CHP. 11. BIOTECHNOLOGY: PRINCIPLES AND PROCESSES	
Boyer and Cohen created r-DNA in year	1972
No. of restriction enzymes known	>900
No. of bacteria strains from which RENs are known	>230
No. of Type II REN known	>350
No. of different types of recognition sites	100
Discovery of transposons by Barbara McClintock in the year	1948
Nobel prize to Barbara McClintock	1948
Percentage of maize genome as transposons	50%
Length of Alu sequence	300 bases
Plasmid size	1 to >1,000 kbp
DNA of λ phage	48.5 kbp
Length of cos sites	12 bp
Bacteriophages used as vector for DNA fragment length	20 kbp
Time for Bacteriophage replication	15 minutes
Lowest number of plasmid copy per cell	1 – 2
High number of plasmid copy per cell	15 -100
Temperature for heat shock	42°C
PCR developed by Kary Mullis in the year	1983
Nobel prize in chemistry to Kary Mullis	1993
DNA segment size usually amplified	100 – 35,000 bp
Taq polymerase can withstand temp. upto	94°C
PCR – Denaturation temperature	91°C
PCR – Annealing temperature	55°C
PCR – Extension temperature	72°C
Reliability of PCR decrease after ____ cycles	28-30
No. of copies by PCR in 30 cycles	1 billion
Time required for each PCR cycle	3 – 5 minutes
CHP. 12. BIOTECHNOLOGY AND ITS APPLICATIONS	
No. of recombinant therapeutics approved for human use	30
No. of recombinant therapeutics marketed in India	12
First gene therapy to 4 year old girl in	1990
First transgenic cow Rosy in the year	1997
Amount of α – lactalbumin in Rosy's milk	2.4 g/L
First vaccine by Edward Jenner	1796
No. of VARIETIES of rice in India	2,00,000
No. of documented varieties of Basmati in India	27
American company patented Texamati in	1997
Human Genome Project was launched in year	1990

Cost of sequencing per bp	US \$3
Cost of total sequencing	US \$9 billions
Books required to store obtained sequences	3300 books each having 1000 pages each containing 1000 letters
HGP was a ____ year project	13
HGP completed in	2003
Sequencing of (last) chromosome 1 completed in	May 2006
No. of bases in human genome acc. To HGP	3164.7 Million
Length of average gene	3000 bases
Length of dystrophin (longest)	2.4 million bases
Total number of genes estimated at	30,000
Functions are unknown for ____ % of discovered genes	>50%
Percentage of genome which codes for proteins	<2%
No. of genes is chromosome 1 (highest)	2968
No. of genes is chromosome Y (lowest)	231
No. of locations identified having SNPs	1.4 million
Genome of Mycobacterium genitalium (one of smallest)	569 kb
Genome of Haemophilus influenza	1830 kb
Genome of Methanococcus jannaschii	1660 kb
Genome of Yeast	12500 kb
Chromosomes in yeast	16
VNTRs first isolated in the year	1980
Probability of having same set of VNTRs	1 in 300 million
DNA fingerprinting technique by Alec Jeffreys discovered in the year	1984

UNIT 10

CHP. 13. ORGANISMS AND POPULATION



Temperature in polar areas and high altitudes	< 0°C
Temperature in deserts in summer	>50°C
Salinity in inland waters	<5ppt
Salinity in sea	30-35 ppt
Salinity in hypersaline lagoons	>100 ppt

Constant body temperature for humans	37°C
Percentage of conformers	99%
Size of Siberian crane population at Bharatpur wetlands	<10
Size of chlamydomonas population in pond	Millions
r for Norway rat	0.015
r for flour beetle	0.12
r values for human population in India in 1981	0.0205
Percentage of phytophagous insects in known insects	25%
CHP. 14. ECOSYSTEM	
Percentage of PAR in incident solar radiation	50%
Percentage of PAR captured by plants	2-10%
Percentage of sunlight energy available into NPP	1%
Percentage of dry weight of organisms as Carbon	49%
Percentage of global carbon found dissolved in oceans	71%
Amount of carbon fixed in the biosphere through photosynthesis annually	4×10^{13} kg
Robert Constanza average price tag to ecosystem services	US \$ 33 trillion (2x Global GNP)
Global gross national product (GNP)	US \$ 18 Trillion
Percentage of ecosystem services price for soil formation	50%
Percentage of ecosystem services price for recreation	10%
Percentage of ecosystem services price for nutrient cycling	10%
Percentage of ecosystem services price for Climate regulation	6%
Percentage of ecosystem services price for Habitat for wildlife	6%

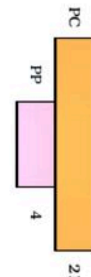


Figure 14.4 (c) Inverted pyramid of biomass-small standing crop of phytoplankton supports large standing crop of zooplankton

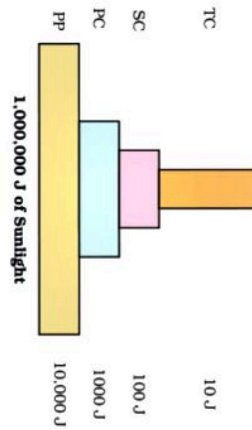


Figure 14.4 (d) An ideal pyramid of energy. Observe that primary producers convert only 1% of the energy in the sunlight available to them into NPP

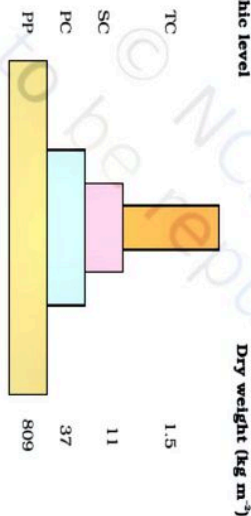


Figure 14.4 (b) Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels

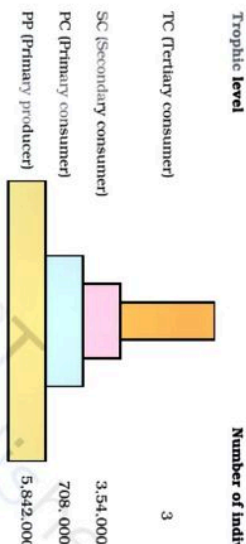


Figure 14.4 (a) Pyramid of numbers in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 millions plants

Species of ants	20,000
Species of beetles	3,00,000
Species of fishes	28,000
Species of orchids	20,000
GENETICALLY DIFFERENT STRAINS of rice in India	50,000
Genetically different varieties of mango in India	1,000
Total number spp. Acc. IUCN (2004)	Slightly more than 1.5 Million
Extreme estimates of no. of species	20 to 50 Million
Robert May estimate of no. of species	7 Million
Percentage of animals in recorded species	>70%
Percentage of plants (including kingdom plantae and Animalia) in recorded species	<22%
Percentage of insects in recorded animal species	70%
Percentage of world land area in India	2.4%
India's share in global species diversity	8.15
No. of megadiversity countries	12
No. of plant species in India	45000
No. of animal species in India	Twice of plants (90000)
Percentage of total species that are recorded (By May's estimates)	22%
No. of plant species yet to be discovered in India	1,00,000
No. of animal species yet to be discovered in India	3,00,000
Latitudinal range of tropics	23.5°N to 23.5°S
No. of bird species in Columbia	1400
No. of bird species in India	1200
No. of bird species in New York (41° N)	105
No. of bird species in Greenland (71° N)	56
Forest in tropical region like Equador has ____ times as many species of vascular plants as a temperate forest of equal area	10
Plant species in Amazon rainforest	40,000
Fish species in Amazon rainforest	3,000
Birds species in Amazon rainforest	1,300
Mammal species in Amazon rainforest	427
Amphibian species in Amazon rainforest	427
Reptiles species in Amazon rainforest	378
Invertebrate species in Amazon rainforest	>1,25,000
Normal Z values	0.1 to 0.2
Z values for entire continents	0.6 to 1.2
Z value for frugivorous birds and mammals in tropical rainforests of different continents	1.15
Amount of extinction caused by colonization of tropical Pacific Islands	>2000 native bird spp.
IUCN Red list (2004) – Total extinction	784 spp.
IUCN Red list (2004) – Vertebrates extinction	338 spp.
IUCN Red list (2004) – Invertebrates extinction	359 spp.

IUCN Red list (2004) – Plant extinction	87 spp.
No. of extinctions witnessed by last twenty years	27 spp.
No. of spp. facing threat of extinction today	15,500
Percentage of birds worldwide threatened	12
Percentage of mammal worldwide threatened	23
Percentage of amphibian worldwide threatened	32
Percentage of gymnosperms worldwide threatened	31
Current extinction rates are estimated to be _____ times faster than pre-human times	100-1000
Amount of spp. which will be wiped out in next 100 years	Half
Past land coverage by tropical rain forest	14%
Present land coverage by tropical rain forest	6%
No. of spp. of cichlid fish extinct due to introduction of Nile Perch in Lake Victoria	200
Percentage of drugs sold worldwide which are derived from plants	25
No. of plant species which contribute to traditional medicines	25,000
Amount of O ₂ contribution by Amazon rainforest	20%
No. of biodiversity hotspots	25 + 9 = 34
Percentage land coverage by biodiversity hotspots	<2%
Strict protection of biodiversity hotspots could reduce ongoing extinction by about	30%
No. of biosphere reserves in India	14
No. of national parks in India	90
No. of wildlife sanctuaries in India	448
Earth summit held in the year	1992
World Summit on Sustainable Development held in the year	2002
No. of countries pledged at World Summit	190
World summit pledge – to achieve significant reduction in biodiversity loss by	2010
CHP. 16. ENVIRONMENTAL ISSUES	
Environment (Protection) Act	1986
Percentage of particulate matter removed by electrostatic precipitator	>99%
Particulate size (diameter) below which it is harmful	2.5 µm
Air (Prevention and Control of Pollution) Act	1981
Amendment to Air act to include noise	1987
Loudness of soundness(jet plane/ rocket) which may damage ear drums	150 dB
Delhi ranked one among ____ most polluted countries in world in 1990s	41
All buses were converted to run on CNG by	End of 2002
Sulphur content in diesel (Euro-III)	350 ppm
Sulphur content in petrol (Euro-III)	150 ppm
Aromatic content in fuel (Euro-III)	42%
Goal - Sulphur content in diesel & petrol	50 ppm
Goal - Aromatic content in fuel	35%
Bharat Stage III (4 wheelers) since	October 2010

Bharat Stage III (3 wheelers) since	October 2010
Bharat Stage III (2 wheelers) since	October 2010
Bharat Stage IV (4 wheelers) since	April 2010
No. of megacities in India	13
Water (Prevention and Control of Pollution) Act	1974
Impurities in water to make it unfit human use	0.1%
Amount of DDT in water -	0.003 ppb
Amount of DDT in Zooplankton -	0.04 ppm
Amount of DDT in Small fish -	0.5 ppm
Amount of DDT in Large fish -	2 ppm
Amount of DDT in Fish-eating birds -	25 ppm
Area of Arcata marshland under FOAM	>60 hectares
Age of Ahmed Khan	57 years old
Ahmed Khan has been producing plastic bags for	20 years
When did Ahmed Khan realize that plastic waste was a problem?	8 years back
Earlier pay of rag pickers	₹ 0.40 / kg
Pay of rag pickers by Khan	₹ 6 / kg
Current membership of Haryana Kisan Welfare Club	5000
Recommended burial depth of nuclear waste	500 m
Current average temperature at Earth surface	15 °C
Average temperature at Earth surface without greenhouse effect	-18°C
Amount of solar radiation reflected back by clouds and gases	$\frac{1}{4}$ th
Amount of solar radiation falls on earth surface	$\frac{1}{2}$
Increase in global temperature in past century	0.6°C
Signing of Montreal protocol	1987
Montreal Protocol effective	1989
Percentage forest loss in tropics	40%
Percentage forest loss in temperate region	1%
Percentage cover of land in India by forest at beginning of 20 th century	30%
Percentage cover of land in India by forest at ending of 20 th century	19.4%
National Forest Policy – Year	1988
National forest policy – percentage coverage by lands – plains	33%
National forest policy – percentage coverage by lands – hills	67%
Amrita Devi Sacrifice	1731
Chipko movement	1974
GOI – JFM	1980s

